

Theorem (Bernoulli's inequality). *Let $x \in \mathbb{R}$ satisfy $x \geq -1$ and let $n \in \mathbb{Z}$ with $n \geq 0$. Then*

$$(1+x)^n \geq 1+nx.$$

Proof. We proceed by induction on the integer exponent n .

1. Base cases.

- $n = 0$. By convention we define $0^0 = 1$; this agrees with the limit $\lim_{y \rightarrow 0^+} y^0 = 1$. Hence

$$(1+x)^0 = 1, \quad 1+0 \cdot x = 1.$$

The inequality holds with equality for every admissible $x \geq -1$.

- $n = 1$. Direct computation gives

$$(1+x)^1 = 1+x, \quad 1+1 \cdot x = 1+x.$$

Equality again holds for all $x \geq -1$.

- $x = -1$. Here $1+x = 0$. For any $n \geq 0$

$$(1+x)^n = 0^n = \begin{cases} 1, & n = 0, \\ 0, & n \geq 1. \end{cases}$$

The right-hand side equals $1+n(-1) = 1-n$. Thus

$$0^n \geq 1-n.$$

This is true because for $n = 0$ both sides are 1, and for $n \geq 1$ the left side is 0 while the right side is at most 0. Equality occurs only when $n = 0$ or $n = 1$.

Hence the statement is valid for the initial exponents and for the special value $x = -1$.

2. Induction hypothesis. Assume that for a fixed integer $k \geq 1$ the inequality

$$(1+x)^k \geq 1+kx \tag{IH}$$

holds for every real $x \geq -1$.

3. Multiplication by the non-negative factor $1+x$. Since $x \geq -1$ we have $1+x \geq 0$. Multiplying both sides of (IH) by this non-negative quantity preserves the direction of the inequality:

$$(1+x)^{k+1} = (1+x)^k(1+x) \geq (1+kx)(1+x). \tag{1}$$

4. Expansion of the right-hand side.

$$(1+kx)(1+x) = 1 + (k+1)x + kx^2. \tag{2}$$

5. Dropping a non-negative term. For any real x we have $x^2 \geq 0$, and because $k \geq 1$ the product kx^2 is non-negative:

$$kx^2 \geq 0. \quad (3)$$

Consequently, from (2) we obtain

$$(1 + kx)(1 + x) \geq 1 + (k + 1)x. \quad (4)$$

6. Inductive step. Combining (1) and (4) yields

$$(1 + x)^{k+1} \geq 1 + (k + 1)x, \quad (1)$$

which is precisely the desired inequality for the exponent $k + 1$.

7. Completion of the induction. The property holds for $n = 0$ and $n = 1$; the inductive step shows that truth for $n = k$ implies truth for $n = k + 1$. By the Principle of Mathematical Induction, the inequality

$$(1 + x)^n \geq 1 + nx$$

holds for every integer $n \geq 0$ and every real $x \geq -1$.

8. Equality cases. Equality in the inductive step can arise in two ways:

1. The discarded non-negative term in (5) vanishes, i.e. $kx^2 = 0$. Since $k \geq 1$, this forces $x = 0$.
2. The factor $1 + x$ that multiplies both sides in (3) is zero, i.e. $x = -1$. In this situation (1) is an equality irrespective of the induction hypothesis.

Collecting all possibilities, equality holds exactly in the following situations:

- $n = 0$ – both sides equal 1 for any $x \geq -1$;
- $n = 1$ – both sides equal $1 + x$ for any $x \geq -1$;
- $x = 0$ – $(1 + 0)^n = 1 = 1 + n \cdot 0$ for any $n \geq 0$;
- $x = -1$ and $n = 1$ – both sides are 0.

For any $n \geq 2$ and any $x \neq 0, -1$ the term kx^2 (and, more generally, the higher-order terms of the binomial expansion) is strictly positive, making the left-hand side strictly larger than $1 + nx$. Hence no other equality cases exist.

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